

Matrix Theory: Lecture 14

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Stochastic Matrices

$x_j^{(t)}$ = probability of state j at time t

p_{ij} = probability of transitioning from state j to state i

$p_{ij} \geq 0$ (probabilities are non-negative)

$\sum_i p_{ij} = 1$ (probability of transitioning from state j to any state)

Probability of state j after transition:

$$x_i^{(t+1)} = p_{i1}x_1^{(t)} + p_{i2}x_2^{(t)} + \dots + p_{in}x_n^{(t)}.$$

In matrix form:

$$X^{(t+1)} = PX^{(t)}.$$

Is there always a stable distribution? $PX = X$? That is, is there a non-negative eigenvector with eigenvalue 1?



Postive Matrices (and Vectors)

Def. A is a **positive matrix** $A > O$ if $a_{ij} > 0$ for all i, j .
 $A > B$ if $A - B > O$.

Similarly A **non-negative matrix** $A \geq O$ if $a_{ij} \geq 0$ for all i, j .

Def. $|A|$ = matrix with elements $|a_{ij}|$.

Inequalites of numbers can be extended:

$$|X + Y| \leq |X| + |Y|, \quad |\alpha X| \leq |\alpha| |X|.$$



Postive Matrices

Theorem 10.14

Let $A > O$ be a positive matrix and $X \geq O$ be non-zero vector. Then

- ① $AX > O$;
- ② A cannot be nilpotent.
- ③ If $AX \geq \alpha X$ for some positive α then $|\lambda| \geq \alpha$ for at least one eigenvalue λ . The inequality is strict if $AX > \alpha X$.
- ④ If Y is any complex vector then $|AY| \leq A|Y|$. The equality is valid iff there exists a non-negative vector P and a real α such $Y = e^{i\alpha}P$.



Positive Matrices

Lemma

Suppose $A > 0$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

If $\rho = \max_i |\lambda_i|$ and $AY = \lambda_k Y$ with $|\lambda_k| = \rho$ then

$$A|Y| = \rho|Y| \quad \text{and} \quad |Y| > 0.$$



Perron Theorem

Theorem 10.14 (Perron)

Let $A > O$ be a positive matrix. Then

- A has a positive eigenvalue $\rho > 0$.
- The corresponding eigenvector X is positive as well and under condition $\sum x_i = 1$ it is unique (and is called **Perron vector**).
- For any other eigenvalue λ (real or complex) one have $|\lambda| < \rho$.
- There are no non-negative eigenvectors other than positive multiplies of the Perron vector.

